A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 39

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B	1
01.2	**1 mark each** 1 mark for each of the 2 points below. **1** Number all possible sampling sites (eg on a grid map) **1** Select the sites using a random number generator / table – (regular shape: grid coordinates from random numbers) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **random** → positions chosen using random numbers so the choice is not biased **Do not credit** • 'pick places at random by eye' - positions have to be generated with random numbers to remove bias	2

Do not write outside the box

Question	Marking guidance	Marks
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** Sites may be chosen for convenience or because they appear to support (or dismiss) the hypothesis **1** So the sample does not represent the whole area and conclusions are invalid **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 391-393; AQA spec 3.7.1.	
02.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** 38.0 / 0.002 **1** = 19 000 (19 000 times / 1.9×10^4) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (times) is already printed on the answer line. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • A bare wrong answer with no working. • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	2
02.2	1 mark for the point below. **1** Biomagnification (accept bioaccumulation as part of the process only if biomagnification is also implied by increase up the food chain) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Biomagnification** → concentration rising at each step along a food chain • **bioaccumulation** → a pollutant building up inside one organism over time **Do not credit** • answers that swap the two terms - bioaccumulation is the build-up inside one organism, biomagnification is the rise along a food chain • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	1

Do not write outside the box

Question	Marking guidance	Marks
02.3	**1 mark each** 1 mark for each of the 4 points below. **1 Persistence:** the pesticide is not broken down (by light, microbes, enzymes) so it remains in the environment / organisms long enough to pass along the food chain **1 Liposolubility:** the pesticide dissolves in lipids / fat so it is stored in body tissue and not excreted in (water-based) urine **1** Each organism eats many organisms from the level below and retains the pesticide so concentration rises at each trophic level **1** Herons have long lives so accumulate over time **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Persistence** → does not break down / stays in the environment for a long time • **trophic level** → the feeding level in a food chain **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	4
02.4	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Organochlorines killed non-target insects (bees, butterflies) and top predators (birds of prey, otters) through biomagnification **1** They persist for decades so the damage continued long after use **1** Use in houses (on walls / nets) does not enter food chains / no runoff into water, so does not biomagnify **1** Benefit of preventing malaria deaths outweighs the risk (risk:benefit) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biomagnification** → concentration rising at each step along a food chain **Do not credit** • answers that swap the two terms - bioaccumulation is the build-up inside one organism, biomagnification is the rise along a food chain • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 263-264, 266; AQA spec 3.4.3.2.8.	
03.1	**1 mark each** 1 mark for each of the 2 points below. **1** pH 5.6 **1** Atmospheric carbon dioxide dissolves in rainwater to form (weak) carbonic acid **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen	2

Do not write outside the box

Question	Marking guidance	Marks
03.2	**1 mark each** 1 mark for each of the 4 points below. **1** Limestone bedrock (calcium carbonate) is alkaline / basic and neutralises the acids as water percolates through it, so the lake pH stays high (7.2) **1** Granite does not neutralise acids so lake water in A is acidic (4.6) **1** Low pH mobilises / dissolves aluminium ions from the soil and rock (0.42 mg dm^{-3} in A) which are toxic to fish (damage gills) **1** Acids also damage fish eggs / gills (exposed living cells) and dissolve the exoskeletons of invertebrate food – lichens and other food organisms lost **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
03.3	**1 mark each** 1 mark for each of the 2 points below. **1 Corrodes metals:** railway lines, railings, pipes, pylons, overhead power lines **1** Dissolves / weakens limestone and marble buildings and statues (eg York Minster) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen	2
03.4	**1 mark each** 1 mark for each of the 2 points below. **1** Sulfur dioxide is oxidised (by ozone / other oxidants) to sulfur trioxide **1** Sulfur trioxide dissolves in water droplets to form sulfuric acid (SO2 alone gives the weaker sulfurous acid) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 250-252; AQA spec 3.4.3.2.2.	

Do not write outside the box

Question	Marking guidance	Marks
04.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2 High-level:** used fuel rods / fission products separated in reprocessing – vitrification in molten glass in sealed stainless steel containers, surrounded by concrete, air-cooled to remove decay heat (Sellafield) **2 Intermediate:** fuel cladding / metal tubes around fuel rods, filters from reprocessing – mixed with cement (encapsulated) in stainless steel drums, stored in concrete vaults **2 Low-level solid:** contaminated clothing, equipment – sealed in plastic bags inside steel drums / containers in a concrete-lined, monitored landfill (Drigg) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **vitrification** → sealing waste into glass **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	6
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** Radioactive decay of the concentrated fission products generates heat which could damage containers **1** Isotopes with long half-lives (plutonium 24 400 years) remain hazardous for thousands of years **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
04.3	**1 mark each** 1 mark for each of the 2 points below. **1 Liquids:** filtered and passed through ion-exchange (sands / resins) to remove radionuclides, then discharged (monitored by CPA / CGM) **1 Gases:** filtered (HEPA / charcoal filters) to remove particles and iodine before release **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 303; AQA spec 3.4.3.2.14.	
05.1	**1 mark each** 1 mark for each of the 2 points below. **1** Molten iron from the blast furnace is converted directly to steel on the same site **1** So it does not cool and need re-melting – transport energy between sites is also avoided – waste heat from one process used by another (eg Kalundborg) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** Collecting and processing small dispersed quantities can use more energy in transport than is saved **1** Mass production and large-scale mining have economies of scale – some materials (glass) take more energy to make than plastic but can be refilled many times **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **recycling** → reprocessing a material to make something new **Do not credit** • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new	2
05.3	**2 marks per point** 2 marks for each of the 2 points below; award 1 mark where a point is only partly made. **2** Transmitting at high voltage means low current so less energy is lost as heat in the cables (loss proportional to current squared) – overhead cables cheaper than underground which need cooling **2** ICT / smart grids co-ordinate real-time data on demand and supply from many sources, switching generation, storage (pumped storage, batteries) and flexible loads so surplus is not wasted and inefficient peaking plants are avoided **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **pumped storage** → water pumped uphill when demand is low and released to generate at peak	4
05.4	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Proximity to the grid / demand to reduce transmission losses and cable costs **1** Access to cooling water for thermal stations **1** Resource availability (wind, tidal range, fuel deliveries) **1** Land availability and planning / public acceptance – stable geology **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 234-235; AQA spec 3.3.4.2.3.	

Do not write outside the box

Question	Marking guidance	Marks
06.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (68 000 - 7 000) / 68 000 x 100 **1** = 89.7% (accept 90%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** Inflow from the rivers was reduced (sometimes to zero) so less water entered while evaporation continued **1** Evaporation removes pure water leaving the dissolved salts behind, so they became concentrated in a smaller volume **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
06.3	**1 mark each** 1 mark for each of the 4 points below. **1** Salinity rose above the range of tolerance of the freshwater fish species / osmotic stress – pollution by pesticides, fertilisers and industrial waste concentrated in the lake **1** Exposed lake bed of salt and pesticide-contaminated dust blown by wind onto farmland and settlements (health problems, soil salinisation) **1** Loss of wetland habitats and wildlife – local climate more extreme (loss of the moderating effect of the water body) **1** Loss of livelihoods – conflict between the countries sharing the rivers **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **range of tolerance** → the range of conditions in which a species can survive • **salinisation** → salt building up in the soil / soil becoming salty **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	4

Do not write outside the box

Question	Marking guidance	Marks
06.4	**1 mark each** 1 mark for each of the 2 points below. **1** Upstream abstraction for irrigation reduces the flow available downstream **1** Upstream HEP dams hold water in summer for winter generation, reducing summer irrigation water downstream – growing populations increase demand **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **abstraction** → water taken out of a river, lake or aquifer • **HEP** → electricity generated from falling water **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 117-118; AQA spec 3.2.2.1-3.2.2.2.	
07.1	**1 mark each** 1 mark for each of the 3 points below. **1** 2020 has higher peak flow in the wet season (32 vs 22 $m^3 s^{-1}$ in March) **1** 2020 has much lower dry-season flow (2 vs 9 $m^3 s^{-1}$ in July / August) **1** Flow is much more variable / seasonal in 2020 – the river almost dries up **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	3
07.2	**1 mark each** 1 mark for each of the 3 points below. **1** In 1990 snow / ice stored wet-season precipitation and released it gradually as meltwater during the dry season, evening out the flow **1** With less ice, precipitation now falls as rain / melts immediately and runs off quickly, increasing wet-season flow and flood risk **1** Little stored ice remains to melt in the dry season so flow falls very low **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	3

Do not write outside the box

Question	Marking guidance	Marks
07.3	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Water shortages for irrigation / drinking / HEP in the dry season **1** Flooding in the wet season **1** Sudden floods from ice-lake outbursts (glacial lake outburst floods) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **HEP** → electricity generated from falling water **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
07.4	**1 mark each** 1 mark for each of the 2 points below. **1** Surface meltwater flows down cracks to the base, lubricating the ice so it slides faster – if this exceeds the faster melting at the front, the front advances **1** But if snowfall at the source has not increased, total volume falls – both retreating and extending glaciers can indicate warming **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 94-96; AQA spec 3.2.1.2.	
08.1	1 mark for the point below. **1** As temperature increases the maximum dissolved oxygen concentration decreases (non-linear: falls faster at low temperatures) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
08.2	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** About 39 °C (accept 38-40) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (°C) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1

Do not write outside the box

Question	Marking guidance	Marks
08.3	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** 9.1 - 7.9 (value at 27 °C, accept 7.8-8.0) **1** = 1.2 mg dm^{-3} (accept 1.1-1.3) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (mg dm^{-3}) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
08.4	**1 mark each** 1 mark for each of the 2 points below. **1** Figure 1 shows the maximum / saturation values – processes that remove oxygen (aerobic decomposition of organic matter / respiration) can give values below the line **1** Higher temperature increases the metabolic rate of organisms and the rate of bacterial decomposition so oxygen is used faster – values above the line are impossible **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
08.5	**Any 3 of 6** 1 mark each for any 3 of the 6 points below, in any order, maximum 3. **1** Increased metabolic rate so more energy used / less surplus for growth **1** Eggs develop faster / hatch earlier (out of step with food supply) **1** Non-indigenous warm-water species may thrive and out-compete indigenous species **1** Toxicity of other pollutants is greater as toxins are metabolised faster **1** Reduced resistance to disease **1** Proteins / enzymes denatured if temperature is very high (above about 40 °C) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
08.6	1 mark for the point below. **1** Hot water is sprayed from a shower floor in a tall hollow tower – air absorbs the heat and rises, drawing in more air (updraught), so the water returned is cooler (and saturated with oxygen) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 256-257; AQA spec 3.4.3.2.6.	

Do not write outside the box

Question	Marking guidance	Marks
09.1	**1 mark each** 1 mark for each of the 3 points below. **1** 1 nitrogen fixation – 2 nitrification **1** 3 denitrification – 4 ammonification **1** 5 ionisation – 6 leaching (3 marks for all six, 2 for four or five, 1 for two or three) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **nitrogen fixation** → nitrogen gas converted into ammonium or other nitrogen compounds • **nitrification** → ammonium oxidised to nitrite and then nitrate • **denitrification** → nitrate converted back to nitrogen gas	3
09.2	**1 mark each** 1 mark for each of the 3 points below. **1** Drainage makes the soil aerobic **1** Aerobic nitrifying bacteria (and nitrogen fixers) increase so more ammonium is converted to nitrate **1** Anaerobic denitrifying bacteria decline so less nitrate is lost as nitrogen gas **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **waterlogged** → soil pores filled with water so there is no air • **nitrifying** → ammonium oxidised to nitrite and then nitrate • **denitrifying** → nitrate converted back to nitrogen gas	3
09.3	**1 mark each** 1 mark for each of the 2 points below. **1** Nitrogen gas is very unreactive so most organisms cannot use it directly **1** Only nitrogen fixation (bacteria, lightning, Haber process) converts it to usable ammonium / nitrate – the cycle recycles a small usable amount **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **nitrogen fixation** → nitrogen gas converted into ammonium or other nitrogen compounds	2
09.4	**1 mark each** 1 mark for each of the 2 points below. **1** Rhizobium bacteria in their root nodules fix atmospheric nitrogen into ammonium / proteins **1** When the plants die / are ploughed in, decomposition and nitrification release nitrates for the following crop **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **nitrification** → ammonium oxidised to nitrite and then nitrate	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 152-154; AQA spec 3.2.4.3.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Context:** Lasky's principle - vast quantities exist at low grade – reserves vs resource – COOG affected by price, technology, energy costs • **Improved exploration:** higher resolution multispectral satellite sensors, IR spectroscopy, gravimetry, magnetometry, seismic, resistivity, portable field analysis, targeted trial drilling • Mechanisation allowing deeper underground mines and larger open-cast mines • **Low-energy extraction of low-grade ores:** bioleaching with acidophilic bacteria and fungi, phytomining, iron displacement, bacterial adsorption of rare earths, polymer adsorption from seawater • **Previously inaccessible deposits:** polymetallic nodules, deep sea sediments, Antarctica (currently protected) - environmental and legal issues • **Extending use:** recycling (advantages, difficulties), Cradle to Cradle design, product lifetime extension, substitution of scarce materials, circular economy • **Evaluation:** environmental impacts vs benefits – energy costs – economics – which strategies are most sustainable – conclusion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **adsorption** → nutrient ions sticking to the surfaces of clay and humus • **recycling** → reprocessing a material to make something new	9
	• **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new	
	Model answer - Level 3 (8 marks)	

Do not write outside the box

Question	Marking guidance	Marks
	Lasky's principle means that as grade falls, the quantity of metal available rises enormously, so depletion of high-grade deposits does not mean running out - it means the cut-off ore grade must fall, which depends on price, technology and energy costs. Better exploration finds what remains: high-resolution multispectral and infrared satellite sensors, gravimetry, magnetometry, seismic and resistivity surveys and portable field analysers narrow the search so expensive trial drilling is targeted. Mechanisation allows deeper underground mines and larger open-cast pits, and previously inaccessible deposits - polymetallic nodules and deep-sea sediments, and Antarctica if its protection were ever lifted - become possible, with serious environmental and legal problems.	
	The most sustainable route to low-grade ores is low-energy extraction: bioleaching with acidophilic bacteria and fungi, phytomining, iron displacement of copper, bacterial adsorption of rare earths and polymer adsorption from seawater all avoid the energy of crushing and smelting vast tonnages. Equally important is extending the use of metal already mined: recycling, Cradle-to-Cradle design so products can be disassembled, longer product lifetimes and substitution of scarce metals - the circular economy approach.	
	Each strategy has costs: deep-sea mining and Antarctica risk unique habitats, deeper mines need more energy, recycling is limited by dispersal and mixed alloys. The strategies that mimic natural low-temperature processes and keep metals in use are the most sustainable, and exploration and new mining should be the last resort rather than the first.	
	Weaker answer - Level 1 (3 marks)	
	When high grade ores run out we can mine lower grade ores instead, but this uses more energy. Better technology like satellites can find new deposits. We could mine the deep sea or Antarctica. Bacteria can be used to extract metals from low grade ore. Recycling metals means we do not need to mine as much. These strategies will help secure supplies of metals in the future.	
	What makes the difference	
	• Level 1 names most strategy types but Level 3 explains each (what the surveys measure, why low-temperature extraction matters, what Lasky's principle implies).	
	• Level 3 weighs environmental and energy costs and ranks the strategies; Level 1 offers no evaluation.	
	• Framing the answer with the concept of cut-off ore grade shows understanding rather than recall.	
10.2	1 mark for the point below. **1** All of the material that exists in the lithosphere, including that which will never be technologically or economically exploitable **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 130-147; AQA spec 3.2.3.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss the strategies that can be used to provide sustainable supplies of water for a growing population.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Context:** uneven distribution of water, growing population and per capita demand, unsustainable exploitation of rivers and aquifers, hydrological cycle links • **New sources:** rainwater collection, rivers (discharge, fluctuations, contaminants), reservoirs and site selection, estuary barrages, desalination, unexploited aquifers • **Management:** artificial aquifer recharge, river-regulation reservoirs, inter-basin transfers, afforestation • **Conservation:** low-water appliances, xeriscaping, drip irrigation, grey water, leak reduction, metering – pollution control to protect sources • **Treatment:** matching quality to use – screens, sedimentation, aeration, flocculation, filtration, sterilisation, pH adjustment, activated carbon, reverse osmosis • Evaluation of costs, energy, environmental impacts (habitat loss, methane, inter-tidal habitats, conflicts) and justified conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.*	25
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 114-127; AQA spec 3.2.2.4-3.2.2.5.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Evaluate the contribution of new technologies to the monitoring and understanding of environmental change in the physical environment.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • Historical data limitations – proxy data - ice cores, isotope ratios (O-18:O-16, C-14), Mg:Ca in calcite, tree rings, pollen, coral • **Satellite sensors:** visible light imagery, infrared for vegetation and temperature, radar microwaves for wind, wave height, sea level and oil, weather monitoring – passing through cloud • **Sensor carriers:** ROVs / drones, AUVs, balloons, aircraft, animals – data recorders for abiotic factors – GPS tracking of floats and sampling sites – Argo floats and ocean monitoring • Climate models and their assumptions – monitoring atmospheric composition (Mauna Loa), ozone (Dobson units), sea level and ice • Applications to hydrosphere, minerals (remote sensing surveys), soils, pollution monitoring • **Evaluation:** improved coverage and resolution vs cost, calibration, interpretation, remaining uncertainty – justified conclusion	25
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 9, 89-107, 402-407; AQA spec 3.2, 3.7.	

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